

A monotone Boolean construction with average–max sensitivity exponent $2/3$

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Abstract

The Servedio–Tan open problem asks whether every monotone Boolean function f satisfies $\text{AS}(f) = o(\text{S}(f))$ [FHH⁺14, ScTa13]. We construct a sequence of monotone Boolean functions H_k with $\text{S}(H_k) \rightarrow \infty$ and

$$\liminf_{k \rightarrow \infty} \frac{\log \text{AS}(H_k)}{\log \text{S}(H_k)} \geq \frac{2}{3}.$$

This gives an explicit monotone family with exponent $2/3$ for the Servedio–Tan average-vs-max sensitivity problem, improving the previously recorded lower-bound exponent $\alpha_{\text{OS}} = \log_2(6 - 2\sqrt{5}) \approx 0.6115$ of O’Donnell and Servedio [OS07].

1 Introduction

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a monotone Boolean function. Two standard complexity measures of f are its average sensitivity (also called total influence) $\text{AS}(f) = \text{Inf}_{1/2}(f)$, the average over a uniform input of the number of pivotal coordinates, and its (maximum) sensitivity $\text{S}(f)$, the corresponding worst-case quantity. Trivially $\text{AS}(f) \leq \text{S}(f)$, and Servedio and Tan (recorded under the heading *Average versus Max Sensitivity for Monotone Functions* in [FHH⁺14]) asked whether monotone functions in fact satisfy

$$\text{AS}(f) = o(\text{S}(f)) \quad \text{as } \text{S}(f) \rightarrow \infty.$$

In the more explicit form stated by Scheder and Tan [ScTa13], the conjecture predicts absolute constants $K > 0$ and $\delta < 1$ such that $\text{AS}(f) \leq K \cdot \text{S}(f)^\delta$ for every monotone f . A natural way to track progress is through the exponent

$$\alpha(f_n) = \liminf_{n \rightarrow \infty} \frac{\log \text{AS}(f_n)}{\log \text{S}(f_n)}.$$

The Servedio–Tan question asks whether $\alpha(f_n) < 1$ for every monotone sequence with $\text{S}(f_n) \rightarrow \infty$. In the opposite direction, O’Donnell and Servedio [OS07] gave a recursive construction achieving

$$\text{AS}(f_n) \geq \text{S}(f_n)^{\alpha_{\text{OS}} - o(1)}, \quad \alpha_{\text{OS}} = \log_2(6 - 2\sqrt{5}) \approx 0.6115,$$

by composing a four-variable base gadget. As noted in [ScTa13], this was the previously recorded lower bound on α for monotone functions in the literature.

Remark 1. As a side observation of this note, a more modest improvement over the O’Donnell–Servedio exponent is also obtained from the Fano-plane line-containment function. This monotone function has sensitivity 3 and average sensitivity $63/32$, so recursive composition gives exponent

$$\frac{\log(63/32)}{\log 3} \approx 0.6166,$$

slightly larger than $\alpha_{\text{OS}} \approx 0.6115$. We do not develop this finite gadget further: the main construction below already gives the qualitatively stronger exponent $2/3$.

Our main result is the following.

Theorem 2. *There is a sequence of monotone Boolean functions H_k with $S(H_k) \rightarrow \infty$ such that*

$$\liminf_{k \rightarrow \infty} \frac{\log \text{AS}(H_k)}{\log S(H_k)} \geq \frac{2}{3}.$$

The argument has three layers. For each prime Q we build a monotone DNF $R_{p,Q}$ over the variable set $\mathbb{F}_Q \times \mathbb{F}_Q$ whose terms are graphs of polynomials of degree less than d for a suitable $d \leq Q$. Sharp first- and second-moment bounds on the number of satisfied terms give, via Paley–Zygmund, a p -biased influence of order Q on a fixed interval of biases. Combining two such blocks — an outer block T at the bias induced by the complementary block, composed with the dual of an inner block B at bias $1/2$ — yields a monotone function with $\text{Inf} \geq cQ^2$ and $S \leq Q^3$. Letting Q range over an unbounded sequence of primes, the resulting sequence satisfies $\liminf \log \text{AS} / \log S \geq 2/3$.

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2 Definitions and elementary composition facts

All Boolean functions are written in 0/1 notation. If V is a finite set, we identify an input $x \in \{0, 1\}^V$ with the set

$$X = \{v \in V : x_v = 1\}.$$

A function $f : \{0, 1\}^V \rightarrow \{0, 1\}$ is monotone if $X \subseteq Y$ implies $f(X) \leq f(Y)$.

For $p \in (0, 1)$, let $X \sim \mu_p$ denote the random subset of V in which each element is included independently with probability p . The p -biased total influence is

$$\text{Inf}_p(f) = \sum_{v \in V} \mathbb{P}_{X \sim \mu_p}[f(X) \neq f(X \Delta \{v\})].$$

The uniform total influence is the average sensitivity:

$$\text{AS}(f) = \text{Inf}_{1/2}(f).$$

For an input X , let $S(f, X)$ be the number of coordinates whose flip changes the value of f , and put

$$S(f) = \max_X S(f, X).$$

For monotone functions it is useful to separate the two sides:

$$S_0(f) = \max_{f(X)=0} |\{v \notin X : f(X \cup \{v\}) = 1\}|,$$

$$S_1(f) = \max_{f(X)=1} |\{v \in X : f(X \setminus \{v\}) = 0\}|.$$

Then

$$S(f) = \max\{S_0(f), S_1(f)\}.$$

The monotone dual of f is

$$f^\dagger(X) = 1 - f(V \setminus X).$$

Then

$$S_0(f^\dagger) = S_1(f), \quad S_1(f^\dagger) = S_0(f),$$

and

$$\text{Inf}_p(f^\dagger) = \text{Inf}_{1-p}(f).$$

If $f : \{0, 1\}^r \rightarrow \{0, 1\}$ and $g : \{0, 1\}^m \rightarrow \{0, 1\}$, their block composition $f \circ g$ is obtained by replacing each input bit of f by an independent copy of g . If

$$q = \mathbb{P}[g(X) = 1]$$

under the uniform measure, then

$$\text{Inf}(f \circ g) = \text{Inf}_q(f) \text{Inf}(g). \tag{1}$$

Indeed, a variable inside the i -th copy of g is pivotal exactly when it is pivotal for that copy of g , and the i -th coordinate of the outer function is pivotal at the vector of block outputs. The other block outputs are independent q -biased bits, and summing over all variables gives (1).

For monotone functions,

$$S_0(f \circ g) \leq S_0(f) S_0(g), \quad S_1(f \circ g) \leq S_1(f) S_1(g). \tag{2}$$

For example, at a zero-input of $f \circ g$, only upward-pivotal coordinates of the outer function can matter, and each corresponding block contributes at most $S_0(g)$ upward-pivotal inner variables. The one-side case is identical.

3 The polynomial-graph DNF block

Fix a prime Q , integers $L \leq Q$ and $d \leq L$, and a subset $E \subseteq \mathbb{F}_Q$ with $|E| = L$. The variable set is

$$V = E \times \mathbb{F}_Q.$$

For every polynomial $P \in \mathbb{F}_Q[z]$ with $\deg P < d$, its graph over E is

$$\Gamma(P) = \{(t, P(t)) : t \in E\} \subseteq V,$$

so $|\Gamma(P)| = L$. The polynomial-graph DNF is

$$R(X) = 1 \iff \Gamma(P) \subseteq X \text{ for some } P \in \mathbb{F}_Q[z], \deg P < d.$$

This is a monotone DNF whose terms all have size L .

Proposition 3 (Algebraic block). *Fix constants $0 < a < b < 1$. There are constants $c_0, c_1 > 0$ and a prime threshold Q_0 , depending only on a, b , such that for every prime $Q \geq Q_0$ and every $p \in [a, b]$, one may choose $L \leq Q$, $d < L$, and E so that the resulting block $R_{p,Q}$ satisfies*

$$c_0 \leq \mathbb{P}_{\mu_p}[R_{p,Q} = 1] \leq \frac{1}{4}, \quad \text{Inf}_p(R_{p,Q}) \geq c_1 Q,$$

$$S_1(R_{p,Q}) \leq Q, \quad S_0(R_{p,Q}) \leq Q^2.$$

Proof. Write $A = \log(1/p)$ and $D_a = \exp(a^{-1} - 1)$. Choose an integer K so large that

$$D_a b^{K-1} \leq \frac{1}{4}, \quad b^{K-1} \leq \frac{1}{4}. \quad (3)$$

For sufficiently large Q , define

$$d = \left\lfloor \frac{(Q-K-2)A}{\log Q} \right\rfloor, \quad L = \left\lfloor \frac{d \log Q}{A} + K \right\rfloor.$$

Then $Q/2 \leq L \leq Q$ and $0 < d \leq L - 1$ for all sufficiently large Q . Choose any $E \subseteq \mathbb{F}_Q$ with $|E| = L$; the DNF $R_{p,Q}$ is defined above.

Sensitivity bounds. Every term of the DNF has size $L \leq Q$, so deleting a variable outside a satisfied term cannot destroy that witness; hence

$$S_1(R_{p,Q}) \leq L \leq Q.$$

The total number of variables is $|V| = LQ \leq Q^2$, so trivially

$$S_0(R_{p,Q}) \leq |V| \leq Q^2.$$

Density bounds. Let $Z(X)$ be the number of satisfied polynomial-graph terms. There are Q^d polynomials of degree less than d , each with a graph of size L , so

$$\lambda := \mathbb{E}_{\mu_p} Z = Q^d p^L.$$

By the choice of L , $L - (d \log Q)/A \in [K - 1, K]$, so

$$a^K \leq \lambda \leq b^{K-1}. \quad (4)$$

For distinct polynomials P, R of degree less than d , the graphs intersect in $|\Gamma(P) \cap \Gamma(R)| = |\{t \in E : (P - R)(t) = 0\}|$ points, the root set of $S := P - R$ inside E ; since $\deg S < d$ and $S \neq 0$, this set has size at most $d - 1$. Substituting and reindexing R as $P + S$,

$$\mathbb{E}_{\mu_p}[Z(Z - 1)] = \sum_P \sum_{R \neq P} p^{|\Gamma(P) \cup \Gamma(R)|} = Q^d p^{2L} \sum_{\substack{S \neq 0 \\ \deg S < d}} p^{-z(S)},$$

where $z(S) := |\{t \in E : S(t) = 0\}|$. Setting $\theta = p^{-1} - 1$ and expanding

$$p^{-z(S)} = (1 + \theta)^{z(S)} = \sum_{T \subseteq \{t \in E : S(t) = 0\}} \theta^{|T|},$$

the inner sum is bounded by

$$\sum_{\substack{S \neq 0 \\ \deg S < d}} p^{-z(S)} \leq \sum_{j=0}^{d-1} \binom{L}{j} \theta^j Q^{d-j},$$

since for each $T \subseteq E$ with $|T| = j < d$ the nonzero polynomials of degree less than d vanishing on T form a set of size at most Q^{d-j} , and any such S has at most $d-1$ roots so $j \leq d-1$. Bounding the binomial sum by extending it,

$$\sum_{j=0}^{d-1} \binom{L}{j} \theta^j Q^{d-j} \leq Q^d (1 + \frac{\theta}{Q})^L \leq Q^d \exp(\theta L / Q) \leq Q^d D_a,$$

using $L \leq Q$ and $\theta \leq a^{-1} - 1$. Therefore

$$\mathbb{E}_{\mu_p}[Z(Z-1)] \leq D_a \lambda^2.$$

Combining with (4) and (3), $\mathbb{E}[Z(Z-1)] \leq \frac{1}{4}\lambda$. Since Z is integer-valued,

$$\mathbb{P}_{\mu_p}[Z = 1] \geq \mathbb{E}Z - \mathbb{E}[Z(Z-1)] \geq \frac{3}{4}\lambda \geq \frac{3}{4}a^K,$$

and Markov's inequality gives

$$\mathbb{P}_{\mu_p}[R_{p,Q} = 1] = \mathbb{P}_{\mu_p}[Z \geq 1] \leq \mathbb{E}Z = \lambda \leq b^{K-1} \leq \frac{1}{4}.$$

Set $c_0 := \frac{3}{4}a^K$.

Influence. On the event $Z = 1$ there is a unique satisfied graph $\Gamma(P)$; each of its L variables is pivotal at X . Therefore

$$\text{Inf}_p(R_{p,Q}) \geq L \mathbb{P}_{\mu_p}[Z = 1] \geq \frac{Q}{2} \cdot \frac{3}{4}a^K \geq c_1 Q$$

with $c_1 := \frac{3}{8}a^K$. □

4 Two-block composition

Apply Proposition 3 on an interval containing $1/2$, for instance $[1/2, 2/3]$. Let $B_Q := R_{1/2, Q}$. Then, for absolute constants $\alpha, c, C > 0$,

$$\alpha \leq \mathbb{P}[B_Q = 1] \leq \frac{1}{4}, \quad \text{Inf}(B_Q) \geq cQ, \quad (5)$$

$$S_1(B_Q) \leq Q, \quad S_0(B_Q) \leq Q^2. \quad (6)$$

After shrinking α if necessary, assume $\alpha < 1/4$. Set

$$G_Q := B_Q^\dagger, \quad \theta_Q := \mathbb{P}[G_Q = 1] = 1 - \mathbb{P}[B_Q = 1] \in [\frac{3}{4}, 1 - \alpha].$$

By the duality identities,

$$\text{Inf}(G_Q) = \text{Inf}(B_Q) \geq cQ, \quad S_0(G_Q) \leq Q, \quad S_1(G_Q) \leq Q^2.$$

Now apply Proposition 3 again on the fixed interval $[\frac{3}{4}, 1 - \alpha]$ (which lies inside $(0, 1)$). For each sufficiently large prime Q , let $T_Q := R_{\theta_Q, Q}$. Then $\text{Inf}_{\theta_Q}(T_Q) \geq c'Q$, $S_1(T_Q) \leq Q$, and $S_0(T_Q) \leq Q^2$, for an absolute constant $c' > 0$.

Define the composed block

$$H_Q := T_Q \circ G_Q,$$

which is monotone. Since the output bias of G_Q is θ_Q , the influence-composition identity (1) gives

$$\text{Inf}(H_Q) = \text{Inf}_{\theta_Q}(T_Q) \text{Inf}(G_Q) \geq c''Q^2, \quad (7)$$

where $c'' := c'c > 0$ is an absolute constant. The sensitivity-composition bounds (2) give

$$S_0(H_Q) \leq S_0(T_Q) S_0(G_Q) \leq Q^2 \cdot Q = Q^3,$$

$$S_1(H_Q) \leq S_1(T_Q) S_1(G_Q) \leq Q \cdot Q^2 = Q^3,$$

so

$$S(H_Q) \leq Q^3. \quad (8)$$

5 The exponent $2/3$

We now read off the asymptotic exponent.

Proof of Theorem 2. Let $(Q_k)_{k \geq 1}$ be a sequence of primes with $Q_k \rightarrow \infty$, and let $H_k := H_{Q_k}$ be the composed block constructed in Section 4. By (7) and (8),

$$\text{AS}(H_k) \geq c'' Q_k^2, \quad S(H_k) \leq Q_k^3.$$

The universal bound $\text{AS}(f) \leq S(f)$ (the pointwise sensitivity is at most its maximum) gives $S(H_k) \geq \text{AS}(H_k) \geq c'' Q_k^2$, so $S(H_k) \rightarrow \infty$. For k large enough that $S(H_k) > 1$,

$$\frac{\log \text{AS}(H_k)}{\log S(H_k)} \geq \frac{2 \log Q_k + \log c''}{3 \log Q_k} = \frac{2}{3} + \frac{\log c''}{3 \log Q_k}.$$

Taking $\liminf$ as $k \rightarrow \infty$,

$$\liminf_{k \rightarrow \infty} \frac{\log \text{AS}(H_k)}{\log S(H_k)} \geq \frac{2}{3}.$$

□

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